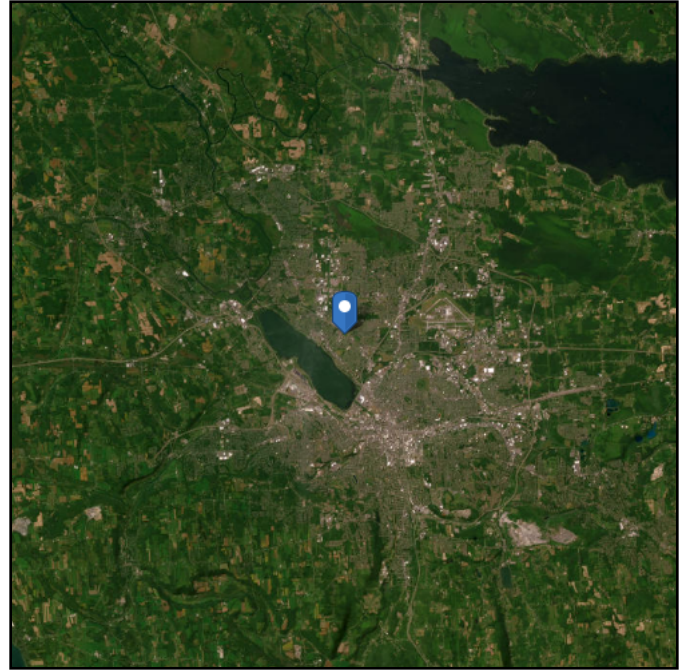
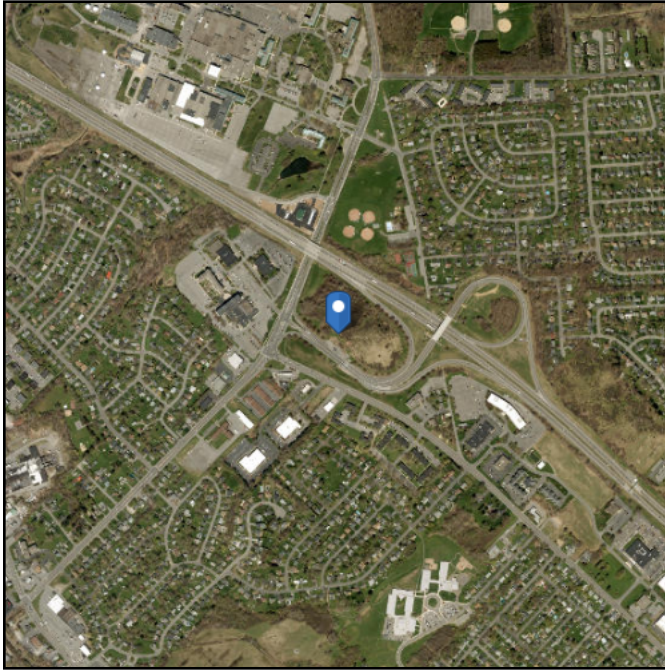


ASCE Hazards Report

Address:
No Address at This Location

Standard: ASCE/SEI 7-22
Risk Category: II
Soil Class: Default

Latitude: 43.101908
Longitude: -76.185071
Elevation: 435.78437137292093 ft
(NAVD 88)



Wind

Results:

Wind Speed	109 Vmph
10-year MRI	74 Vmph
25-year MRI	81 Vmph
50-year MRI	86 Vmph
100-year MRI	92 Vmph
300-year MRI	102 Vmph
700-year MRI	109 Vmph
1,700-year MRI	116 Vmph
3,000-year MRI	121 Vmph
10,000-year MRI	133 Vmph
100,000-year MRI	154 Vmph
1,000,000-year MRI	175 Vmph

Data Source:

Date Accessed:

ASCE/SEI 7-22, Fig. 26.5-1B and Figs. CC.2-1–CC.2-4,
and Section 26.5.2
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Value provided is 3-second gust wind speeds at 33 ft above ground for Exposure C Category, based on linear interpolation between contours. Wind speeds are interpolated in accordance with the 7-22 Standard. Wind speeds correspond to approximately a 7% probability of exceedance in 50 years (annual exceedance probability = 0.00143, MRI = 700 years). Values for 10-year MRI, 25-year MRI, 50-year MRI and 100-year MRI are Service Level wind speeds, all other wind speeds are Ultimate wind speeds.

Site is not in a hurricane-prone region as defined in ASCE/SEI 7-22 Section 26.2.

Site Soil Class:

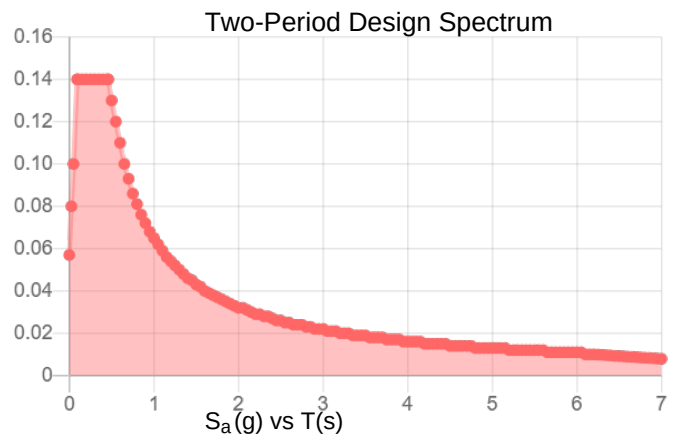
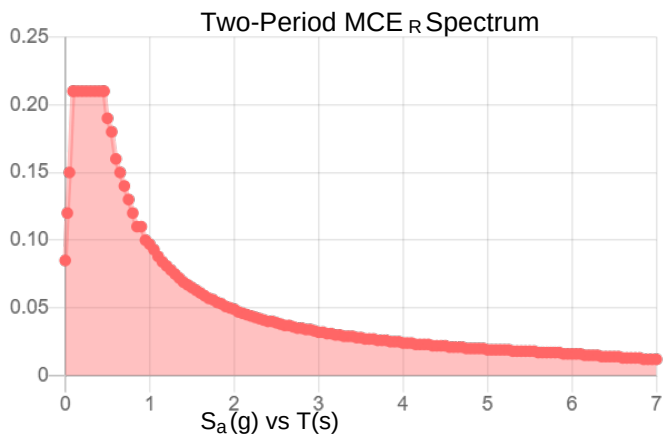
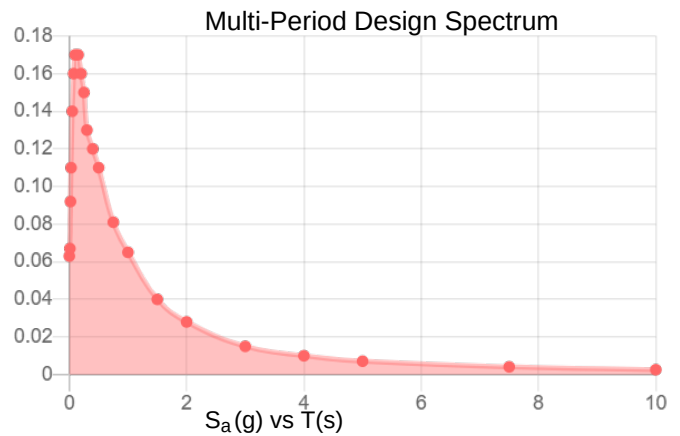
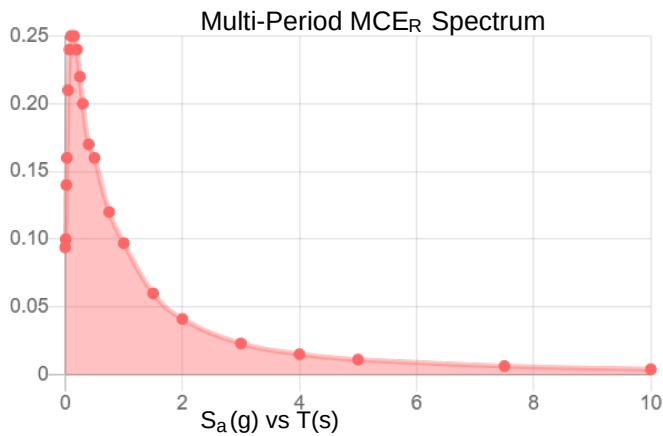
Default

Results:

PGA_M : 0.085
S_{MS} : 0.21
S_{M1} : 0.097
S_{DS} : 0.14
S_{D1} : 0.065

T_L : 6
S_S : 0.17
S₁ : 0.046
V_{S30} : 260

Seismic Design Category: A



MCE_R Vertical Response Spectrum
Vertical ground motion data has not yet been made available by USGS.

Design Vertical Response Spectrum
Vertical ground motion data has not yet been made available by USGS.

Data Accessed:

Wed Jun 25 2025

Date Source:

USGS Seismic Design Maps based on ASCE/SEI 7-22 and ASCE/SEI 7-22 Table 1.5-2. Additional data for site-specific ground motion procedures in accordance with ASCE/SEI 7-22 Ch. 21 are available from USGS.

Results:

Ice Thickness: 1.00 in.
Concurrent Temperature: 5 F
3-s Gust Speed: 44 mph

Data Source:

Standard ASCE/SEI 7-22, Figs. 10-2 through 10-8

Date Accessed:

Wed Jun 25 2025

Ice thicknesses on structures in exposed locations at elevations higher than the surrounding terrain and in valleys and gorges may exceed the mapped values.

Values provided are equivalent radial ice thicknesses due to freezing rain for 250, 500, 1,000, and 1,400-year mean recurrence intervals along with concurrent 3-s gust speeds and concurrent air temperatures. The shading indicates special icing regions, with elevations above 2,100 ft (640 m) in the east, 6,000 ft (1829 m) in the west, and 1,600 ft (488 m) in Alaska, with sparse weather station data for determining design ice loads. In these regions, as well as in regions with complex terrain causing unusual icing conditions and regions where snow or in-cloud icing results in larger loads, the mapped values should be adjusted based on a combination of local historical records and experience, reanalysis data, and numerical weather prediction systems.

Snow

Results:

Ground Snow Load, p_g : 83 lb/ft²
20-year MRI Value: 37.93 lb/ft²
Winter Wind Parameter: 0.55
Mapped Elevation: 404.3 ft

Data Source:

ASCE/SEI 7-22, Figures 7.6-1 and 7.6-2 A-D

Date Accessed:

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Values provided are ground snow loads. In areas designated "case study required," extreme local variations in ground snow loads preclude mapping at this scale. Site-specific case studies are required to establish ground snow loads at elevations not covered.

Snow load values are mapped to a 0.5 mile resolution. This resolution can create a mismatch between the mapped elevation and the site-specific elevation in topographically complex areas. Engineers should consult the local authority having jurisdiction in locations where the reported 'elevation' and 'mapped elevation' differ significantly from each other.

Ground Snow Loads for IRC only, $p_{g(asd)}$: 58.1 lb/ft²

Rain

Results:

15-minute Precipitation Intensity: 5.22 in./h

60-minute Precipitation Intensity: 2.36 in./h

Data Source:

NOAA National Weather Service, Precipitation
Frequency Data Server, Atlas 14([https://
www.nws.noaa.gov/oh/hdsc/](https://www.nws.noaa.gov/oh/hdsc/))

Date Accessed:

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Results:

Flood Zone Categorization:

X (unshaded)

Base Flood Elevation:

Refer to map for local elevations and interpolate according to the Authority Having Jurisdiction.

Data Source:

FEMA National Flood Hazard Layer - Effective Flood Hazard Layer for US, where modernized (<https://msc.fema.gov/portal/search>)
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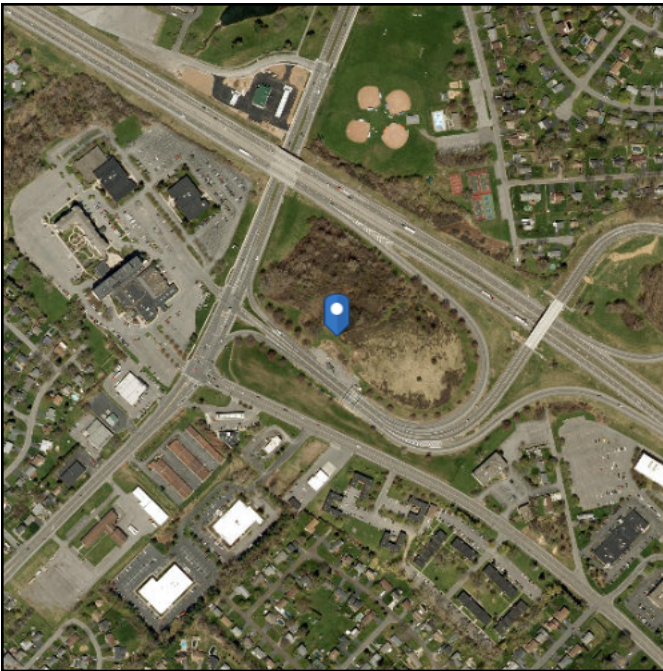
Date Accessed:

FIRM Panel:

If available, download FIRM panel [here](#)

Insurance Study Note:

Download FEMA Flood Insurance Study for this area [here](#)



Tsunami

Results:

Tsunami:

Not in mapped tsunami design zone.

Data Source:

[ASCE Tsunami Design Geodatabase](#)

Date Accessed:

Wed Jun 25 2025

Tornado

Results:

Not Applicable to Risk Category 2

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